

CyberLogix MC2 Motion Controller Documentation

The document describes the Ports and indicators of the MC2 Motion controller

(Revised: 11/04/16)

Updated for FW12 adding of motor stall detection 22/8/15

Version 12

The MC2 Motion controller is a network driven servo motor controller capable of driving brushed or brushless motors

Specs:

Motor supply voltage: 20 to 180V DC

Current: 10 amps continuous 30 amps peak

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Front View of Motion Controller

- | | |
|---|---|
| Encoder Plug and indicators (Orange Lead Connects to this port) | Red Led indicates Encoder wiring Error
Green LED indicates motion move complete |
| Hall Plug and Indicators (Purple Lead connects to this port) | Red LED Indicates Hall wiring error or incorrect brush/brushless setting in software
Red flashing indicates Firmware update mode
Green LED Flashes to indicate motion CPU is OK |
| Status LED Display | See Below for LED Status messages
Decimal Point indicates Drive Enabled |
| Control Wiring Plug and Indicators | Green LED indicates 24V Supply OK
Green LED indicates Enable Input is OK
Green LED indicates Home sensor is ON
Green LED indicates High Limit switch is OK (if limit inputs enabled in settings)
Green LED indicates Low Limit switch is OK (if limit inputs enabled in settings) |
| See Below for correct wiring | |
| Motor Wiring Plug and LED | Green LED Indicates Motor power is OK (Very dim if motor volts is 24v) |

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Status LED Display

Display will scroll around in a circle if all is OK

Or flash a 3 alphanumeric code for status or fault

Status

SLL	Software Low Limit	The drive is at a software limit and will only respond to higher position setpoints
SHL	Software High Limit	The drive is at a software limit and will only respond to lower position setpoints
HLL	Hardware Low Limit	The Low Limit switch is off and drive will only respond to forward motion
HHL	Hardware High Limit	The High Limit switch is off and drive will only respond to Reverse motion

Faults

F01	Invalid hall state on Hall inputs	Check hall wiring or motor hall sensors or that controller is set in correct brush/brushless mode
F02	Encoder Wiring Fault	Check encoder wiring or encoder on motor
F03	Encoder Power Fault	Internal auto reset fuse has tripped due to over current on encoder supply Check encoder wiring or encoder on motor
F04	Position Error limit exceeded	Check for jam on machine and that motor can turn freely, check if trying to drive motor too fast, Check for under voltage on motor or faulty motor or encoder, check current limiting and output limiting in drive
F05	Motor Over current fault	Peak current limit has been reached on servo drive check for faulty wiring or motor or overloaded
F06	Motor Power Fault	The motor supply voltage is either to high or low
F07	Temperature Fault	The drive is overheating ensure adequate ventilation, overloading etc fan force cooling if required
F08	Amp Disabled	Massive over current detected by Drive Amp, check for a short circuit on the motor or wiring or its also Possible for this to happen if the motor output is hard stopped very suddenly
F09	Enable Lost	While the drive was enabled and holding position or moving it lost its enable (Emergency stop) input
F10	Motor Stalled	The motor is not moving while it has full permissible power applied check as per F04
F11		Call Cyberlogix if you see this fault
F12		Call Cyberlogix if you see this fault
F13		Call Cyberlogix if you see this fault

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F14	Comms Fail in software)	Host device communications has timed out (Drive must have host comms every 3 secs, this can be adjusted
F15	Drive Not Setup	Drive has not been setup, send setup message to drive (normally by a reset button in software)
F16	No Address	Drive has not been configured by host device, check communication cables and host device
F99	CPU Not Running	Call Cyberlogix – Unit is in Flash update mode or the CPU has failed

Green indicators on front from top to bottom next to control wiring plug

24v control power indicator

Drive Enabled input

Drive home sensor input

Drive hardwired low limit input (remember to enable limit switches in software)

Drive hardwired High limit input (remember to enable limit switches in software)

Drive Motor Supply LED (Will be very dim on 24v Motor supply and very bright on 180v motor supply!)

Green Control wiring Plug Connections Top to Bottom

- 1 24V Control Power Input (Positive supply) - **with LED indication**
- 2 24V Control Power, internally connected to the terminal above (can be used for inputs below)
- 3 Drive Enable Input - **with LED indication**
- 4 Drive Home Sensor Input - **with LED indication**
- 5 Drive hardwired High limit, input is Fail safe so power to this terminal means its ok to move in positive direction (remember to enable limit switches in software) - **with LED indication**
- 6 Drive hardwired Low limit, input is Fail safe so power to this terminal means its ok to move in positive direction (remember to enable limit switches in software) - **with LED indication**
- 7 0v Return control power, internally connected to the terminal below (can be used for sensors for inputs above)
- 8 0v Return control power supply (Negative return)

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Motor Plug

- 1 High voltage motor supply (Positive supply) - **with LED indication**
- 2 High voltage motor supply return (Negative return)
- 3 U connection to motor (or in brush systems + to Motor)
- 4 V connection to motor (or in brush systems – to motor)
- 5 W connection to motor (no connection in brush motors)
- 6 Motor ground (connected internally to Negative return and also alloy case)



Top View of Motion Controller

Upstream comms towards host controller

Red LED Indicates Comms TX to host

Green LED Indicates Comms RX from host

Downstream comms to further modules or no connection if the last device on the network

Red LED Indicates this is last module on comms link

Green LED Indicates 5V supply on PCB is OK